

Line Maintenance Operational Dashboard

Live app: <https://jfk-ops-live-dashboard-vercel-mirro.vercel.app/>

Executive Summary

The Line Maintenance Operational Dashboard is a low-code, station-tailorable web application that brings daily roster planning, live operational visibility, weather awareness, and resource assignment into one execution surface. It was designed for practical line maintenance decision-making under time pressure, where fragmented systems and static reports often create delays and coordination gaps.

Core value: one operational board for engineers and managers to plan, execute, and rebalance workload in real time.

Problem Statement

- Daily roster files vary by station and format, making consistent interpretation difficult.
- Assignment coordination between certifiers and mechanics is often manual and time-sensitive.
- Operational context (weather, live movement, timeline congestion) is spread across multiple tools.
- Managers need fast visibility of peak workload windows to avoid resource bottlenecks.

Solution Design

1) Interpretative roster ingestion

A Daily Roster CSV upload module acts as the baseline input layer. The ingestion logic interprets roster content and transposes it into a structured operational board, including flight lines and assignment-ready records. This enables compatibility with varying station roster layouts and reduces setup friction.

2) Live execution board

- Flight list with status, A/C registration, A/C type, terminal/gate, ETA/STD, and notes.
- Editable assignment columns for certifier/mechanic with rapid operational updates.
- Column-level reset controls for quick board hygiene during dynamic changes.
- Station weather tile (current + day-part outlook) for tactical awareness.

3) Manager analytics view

- 24-hour Gantt timeline for flight overlap and sequencing visibility.
- Heat-map intensity overlay to identify peak workload periods.
- Supports staffing rebalance decisions and shift handover quality.

Enrichment & Data Resilience

The dashboard uses a layered enrichment architecture to reduce single-source dependency and maintain operational continuity.

- **Primary/secondary feed strategy:** modular provider routing with fallback handling.
- **Cache layer:** retains last-known useful values to reduce blank operational fields.
- **Manual override layer:** allows frontline correction where external feeds are incomplete.
- **Type normalization:** DOC8643-based aircraft type mapping to cleaner, standardized outputs.

Key Capabilities

- Rapid daily deployment from CSV input to actionable roster board.
- Resource assignment controls with practical reset and correction workflows.
- Real-time weather + operational context in one interface.
- Timeline and peak-demand visibility for management decisions.
- Low-code architecture, easy to tailor by station without high licensing overhead.

Practical Use Cases

- **Shift startup:** load roster and publish a consistent board view in minutes.
- **In-day disruption handling:** quickly reassign certifier/mechanic resources.
- **Handover quality:** preserve notes, assignment intent, and timeline clarity.
- **Peak-time planning:** use heat-map/Gantt to pre-position capability before congestion windows.
- **Multi-station scaling:** replicate core app while tailoring parser and assignment defaults locally.

Why this approach works

- **Operationally native:** built around line maintenance workflows, not generic project tooling.
- **Flexible input model:** adaptable to real-world roster variability.
- **Low-cost execution:** web-first, lightweight, and easy to maintain.
- **Human-in-the-loop by design:** automation accelerates decisions, while engineers/managers keep control.

Screenshots

Live tracker temporarily unavailable; using fallback snapshot + schedule.

Airline	Flight	A/C Reg	A/C Type	Terminal	Gate	ETA	STD	Status	Certifier	Mechanic	Notes
JL	JL006/005	-	A350	8	-	1100	1350	Scheduled	Andrew Ibsen	Assign...	-
NH	NH110/109	-	B777	7	-	1130	1405	Scheduled	Andrew Ibsen	Assign...	-
BA	BA117/176	-	B787	8	18C	1217	2110	Scheduled	Andrew Ibsen	Nikolas Dundon	-
BA	BA175/172	-	B787	8	18D	1334	2140	Scheduled	Andrew Ibsen	Nikolas Dundon	-
EI	EI105/104	-	A330	7	-	1455	1800	Scheduled	Rahman Arkan	Fahim Abrar	-
BA	BA173/112	-	B787	8	20	1521	1915	Scheduled	Devran Turegun	Nikolas Dundon	-
QF	QF3/4	-	B787	8	-	1525	1815	Scheduled	Ray Abes	Naresh Dindiall	-
EI	EI111/110	-	A321	7	-	1625	1830	Scheduled	Mark Ferrel	Fahim Abrar	-
IB	IB211/212	EC-OIL	A32X	8	-	1625	1755	Scheduled	Saleh Al Assaf	Assign...	-
ZO	Z0701/702	G-CKWP	B787	7	-	1655	1905	Scheduled	Assign...	Assign...	-
BA	BA177/174	-	B787	8	16	1658	1950	Scheduled	Rahman Arkan	Frank Richmond	-
NZ	NZ2/1	-	B787	1	-	1700	1920	Scheduled	Assign...	Assign...	-

Figure 1: Daily roster execution board. Editable assignment columns, operational status, and structured flight line view for shift control.

Documentation & Examples

Everything you need to Integrate with ADS-B Exchange

 Quick Start Examples

JavaScript

Python

cURL

ADS-B Exchange Javascript Example

JAVASCRIPT

```
// ADS-B Exchange API Example
const response = await fetch('https://www.adsbexchange.com/data/', {
  method: 'GET',
  headers: {
    'Content-Type': 'application/json'
  }
});

const data = await response.json();
console.log(data);
```

Copy

Unlock the power of the ADS-B Exchange API, which provides seamless access to comprehensive real-time and historical data of airborne aircraft. With a robust dataset that covers a wide array of vital flight information, this API is designed for developers, aviation enthusiasts, and organizations requiring detailed aerial insights. Whether you are building flight tracking applications, conducting aviation research, or simply eager to stay informed about ongoing air traffic, the ADS-B Exchange API delivers a reliable solution to tap into an extensive pool of aircraft data, ensuring you stay updated with the latest airborne activity at your fingertips.

Utilizing the ADS-B Exchange API comes with numerous advantages. The key benefits include:

- Real-time tracking of active flights, enhancing situational awareness.
- Access to historical flight data for comprehensive analysis.
- Coverage of a vast number of aircraft worldwide, ensuring extensive data availability.
- User-friendly documentation and support, allowing for quick integration.
- Free access to open-source data, promoting transparency and collaboration in aviation information sharing.

Figure 2: Live context panel showing weather, feed health, and active board state for tactical decision-making.

Access

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